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## Assessment

### Regression Task Analysis Report

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## Abstract

**Purpose:** This report aims to analyze the Gender Development Index (GDI) of the whole world by examining different factors affecting the development of genders.

**Approach:** The dataset used is Gender Development Index 2019 which measures gender gaps in human development achievements. The dataset aligns with the Sustainable Development Goal (SDG) 5, that is Gender Equality. The dataset consists of 14 columns and 189 rows, each row representing a country. The dataset is examined to contain null or duplicate values, performed necessary Exploratory Data Analysis (EDA) and created two different models for predicting the target.

**Key Results:** Both the models were evaluated fairly using Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared. Both the models performed well demonstrating that the dataset was satisfactory for the regression task. The linear regression model had R-squared value of around 0.78 while the KNN regression model had R-squared value of around 0.46.

**Conclusion:** The final regression model performed excellently with R-squared value of 0.778, Root Mean Squared Error of 0.026 and Mean Squared Error of 0.0006. For this dataset, Linear Regression model was way better than KNN model with almost 30 percent performance increase.

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# 1. Introduction

## 1.1. Problem Statement

The goal of this project is to develop a machine learning model to predict the Gender Development Index (GDI) value using key features such as Life Expectancy, Expected years of schooling and Human Development Index of both the genders, male and female.

## 1.2. Dataset

The project uses a dataset, Gender Development Index 2019 available in Kaggle. The dataset consists of 189 rows and 14 columns. Each of the rows represents a country. This dataset consists of key features to identify Development Index, each for both the genders. The dataset aligns with the Sustainable Development Goal 5: Gender Equality as it identifies the development achievements of both the genders.

## 1.3. Objective

The objective of this task is to build two different models for regression tasks that can predict the GDI value of each of the country with the help of other independent variables. After that, both the models need to be evaluated and compared using R-squared values, Mean Squared Error and Root Mean Squared Error to find the best model for the dataset.

## 2. Methodology

### 2.1. Data Preprocessing and Exploratory Data Analysis

Before building the model, it is required to analyze whether the dataset is usable or not. If not, the dataset needs to be made usable. For this, it is important to perform data preprocessing and Exploratory Data Analysis (EDA). Firstly, the dataset was loaded into a data frame. To make the data usable, the data was first analyzed by using different EDA techniques. One of the major issues in the dataset is that it had “..” values instead of NAN values and the independent columns were of type object. To fix this, the “..” values were first converted into NAN. It was also seen that the data had commas, for example 3890.58 was written as 3,890.58 so the commas were also removed. After that the independent variables were converted to float type. The NAN values were then filled with the mean of the respective columns.

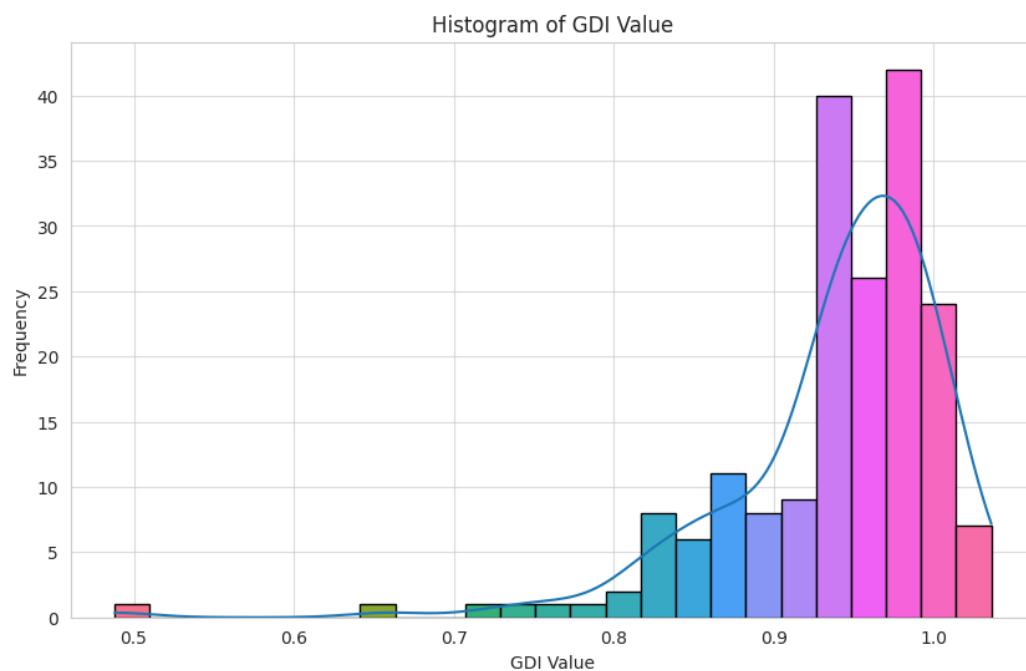


Figure 1Histogram of GDI Values

The above figure is a histogram representing the GDI Value (Target Variable). This illustrates the distribution of the target variable. The shape of the histogram is skewed on the left. This highlights that there is a small number of outliers present in the data.

After making the data usable, it was prepared for the model learning process. Firstly, independent columns from the data were selected for features and the “GDI\_value” column was selected as the target. The data is then split into training and testing set. The data is then scaled using normalization.

## 2.2. Model Building

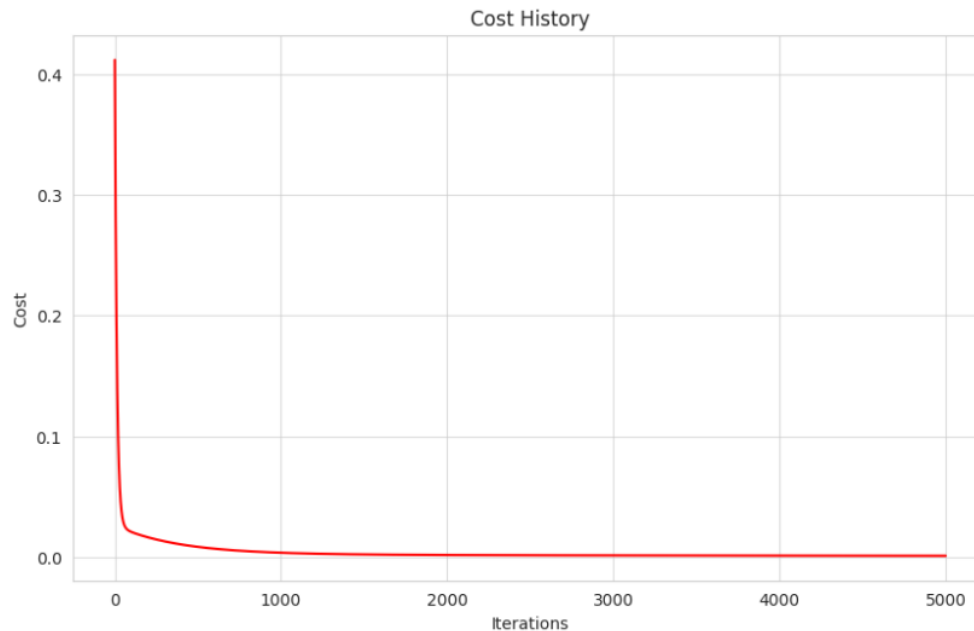
The scaled data is then used to build two different models, one for Linear Regression and one for KNN Regression. For this, different functions were created to make the model reusable and scalable in the future.

### Linear Regression Model

Linear regression model depends on the gradient descent to optimize the parameters weights( $w$ ) and bias( $b$ ). In this model, it is necessary to compute the cost of the model which is the error between predicted and actual values.

To create the model:

- i. Predict  $y$ : To predict the  $y$  we use the  $y = w.x + b$  formula where,  $x$  is the independent variable,  $w$  is the weight,  $b$  is the bias.
- ii. Cost Computation: In linear regression, Mean Squared Error is used to compute the cost between predictions and actual values.
- iii. Gradient Descent: The gradient of the cost is performed and using a learning rate ( $\alpha$ ), the parameters  $w$  and  $b$  are updated. After a fixed number of iterations, the gradient descent stops, and the optimal value of  $w$  and  $b$  is obtained where the cost is minimum.
- iv. Using the optimal value: then the optimal values of  $w$  and  $b$  were used to predict the test set.



*Figure 2 Cost History of Linear Regression Before Feature Selection*

The above graph shows the difference in cost as the number of iterations increases. It can be observed that the cost decreases as the number of iterations increases. The cost descends slowly after a certain number of iterations are reached.

## KNN Regression Model

K Nearest Neighbors (KNN) Regression Model depends on the k number of its neighbors to predict the value. The model finds the nearest neighbors and finds the mean of those neighbors. The mean is taken as the predicted value.

To create the model:

- i. Computing Distance: The distance between the data points and the predicting value is calculated using Euclidean Distance formula.
- ii. Predict KNN: Then using the given value of k, finding the mean of the k nearest neighbors and using it as predictions.

## 2.3. Model Evaluation

After building, the models were evaluated using MSE, RMSE and R-squared values for both test and train set.

### Linear Regression Model

|                  | Train Set | Test Set |
|------------------|-----------|----------|
| <b>MSE</b>       | 0.002435  | 0.001527 |
| <b>RMSE</b>      | 0.049341  | 0.039073 |
| <b>R-Squared</b> | 0.541514  | 0.507032 |

The above table shows the evaluation of Linear Regression Model. The table compares the values of train set and test set. By looking at the R-Squared value, the model is neither overfitting nor underfitting. The initial model was evaluated based on alpha = 0.01 and iteration = 5000.

### KNN Regression Model

|                  | Test Set |
|------------------|----------|
| <b>MSE</b>       | 0.001605 |
| <b>RMSE</b>      | 0.040056 |
| <b>R-Squared</b> | 0.481913 |

The above table shows the evaluation of KNN Regression Model. The table describes the values of test set only as KNN does not have any training phase. The initial model was evaluated based on k = 3.

## 2.4. Hyper-parameter Optimization

A grid search algorithm was used for hyperparameter tuning. To find the best hyperparameter, it was necessary to use grid search instead of random search despite its performance benefits as the model was being highly inconsistent while using random search.

### Linear Regression Model:

For linear regression, the hyperparameters to determine are alpha (learning rate) and iteration (iteration count for gradient descent).

For hyperparameter tuning, the following lists were used.

`alpha_list = [0.01, 0.02, 0.03, 0.04, 0.5, 0.1]`

`iterations_list = [1000, 5000, 10000, 20000]`

After using grid search, the best hyperparameters were:

- i. Alpha = 0.3
- ii. Iterations = 5000

The model had a R-squared value of 0.80.

### KNN Regression Model:

For KNN regression, the hyperparameter to determine is k (number of nearest neighbors to select).

For hyperparameter tuning, the following list was used.

`k_list = [3, 5, 7, 9, 11, 13, 15, 17]`

After using grid search, the best hyperparameter was:

- i. K = 3

The model had a R-squared value of 0.49.

## 2.5. Feature Selection

SelectKBest with  $f_{\text{regression}}$  score function provided by scikit learn was used for feature Selection. The total number of usable features was 10 for regression tasks out of which 7 were selected.

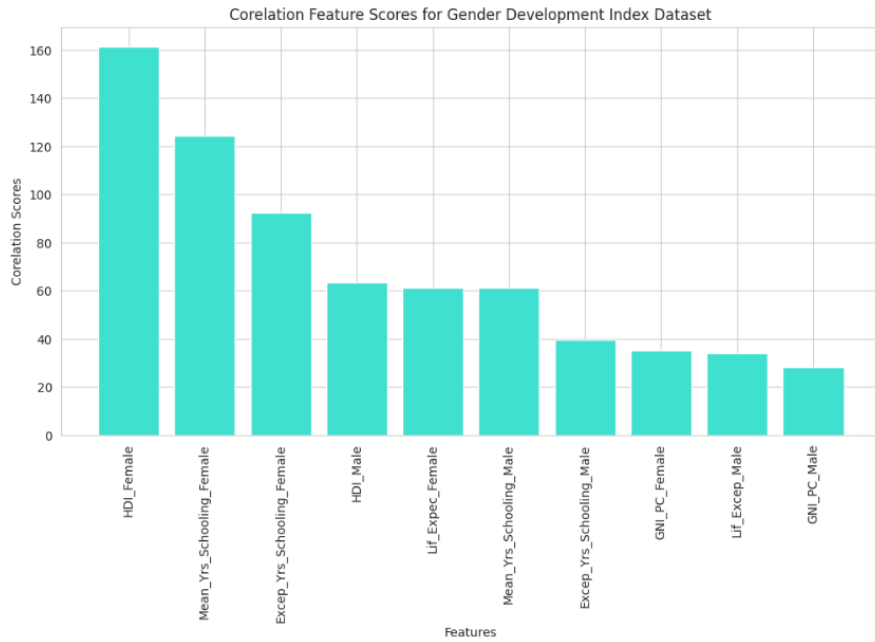


Figure 3Correlation Score of Features

The above figure describes the correlation of all 10 features with the GDI value. The HDI\_Female has the highest relation with the GDI value while GNI\_PC\_Male value has the lowest relation. Even though the relation is low, it is still usable as an independent variable for this regression task.

### Linear Regression After Feature Selection with Hyperparameter Tuning:

|           | Train Set | Test Set |
|-----------|-----------|----------|
| MSE       | 0.001437  | 0.000686 |
| RMSE      | 0.037     | 0.026196 |
| R-Squared | 0.729420  | 0.778411 |

### KNN Regression After Feature Selection with Hyperparameter Tuning:

|           | Test Set |
|-----------|----------|
| MSE       | 0.001655 |
| RMSE      | 0.040682 |
| R-Squared | 0.465608 |

### 3. Conclusion

#### 3.1. Key Findings

After evaluating the models with hyperparameter tuning and feature selection, it was seen that the performance of the model depends on its hyperparameters and its features. There was a high increase in the R-squared value of the Linear Regression after finding a set of the best hyperparameters while there was only a slight increase for KNN Regression task.

#### 3.2. Final Model

For this dataset, Linear Regression Model was used as its final model. After comparing the evaluation of the R-squared value of both the models after feature selection and hyperparameter tuning in test set, Linear Regression had a better score with 0.729 while KNN had a score of 0.465.

#### Building Linear Regression as Final Model:

To build the Linear Regression model, the best hyperparameter was used with feature selection.

Hyperparameters:

- i. Alpha = 0.03,
- ii. Iterations = 5000

Features Selected: 7

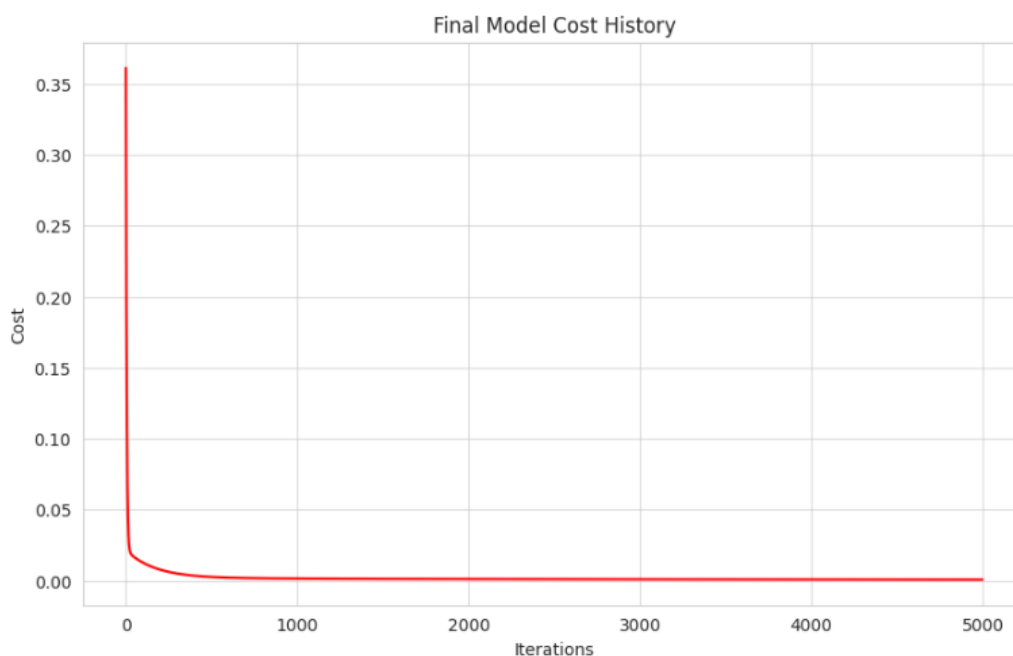


Figure 4 Cost History of Final Model

The figure shows the behavior of the cost function of the final model. The cost decreases as iteration increases. The final cost was 0.000718 from the initial cost of 0.361271.

|                  | <b>Train Set</b> | <b>Test Set</b> |
|------------------|------------------|-----------------|
| <b>MSE</b>       | 0.001437         | 0.000686        |
| <b>RMSE</b>      | 0.037            | 0.026196        |
| <b>R-Squared</b> | 0.729420         | 0.778411        |

### 3.3. Challenges

One of the challenges faced was during data analysis. Since the data had “.” instead of NAN values, it was providing a problem for any manual mathematical computations. The dataset required to convert “.” into NAN values and then fill the NAN values with the mean. Another major challenge faced was the predictions being inconsistent as the hyperparameters changed. To fix this problem, it was necessary to use grid search as random search was having the predictions very inconsistent.

### 3.4. Future Work

The model has been able to predict the values more precisely and accurately than expected. But in future, instead of Grid Search which consumes a lot of computational cost, more optimized hyperparameter tuning algorithms can be explored to obtain optimal hyperparameters for building a better model.

## 4. Discussion

### 4.1. Model Performance

The performance of the models depended on MSE, RMSE and R-Squared Error. Comparing both models, Linear Regression Model performed better than KNN Model with MSE of 0.000686, RMSE of 0.026196 and R-Squared Error of 0.778411 on test sets. Overall, both models performed well for this dataset, but Linear Regression model worked better so it was chosen to build a final model using the selected features and hyperparameters.

### 4.2. Impact of Hyperparameter Tuning and Feature Selection

After building the final model using the obtained set of hyperparameters and features, the performance of the model improved as MSE decreased and RMSE decreased whereas  $r^2$  increased. This means that the model is performing well and fitting the data properly.

### 4.3. Interpretation of Results

The final interpretation for the output obtained from all the models on test set are:

#### For Linear Regression Model:

MSE: 0.001527

RMSE: 0.039073

R-Squared: 0.507032

#### For KNN Regression Model:

MSE: 0.001605

RMSE: 0.040056

R-Squared: 0.481913

### 4.4. Limitations

Due to less data being presented in the dataset, the model may be overfitted due to less data to perform training and testing tasks. The total data present in the data set was 189 and out of them 22 rows of data consisted of missing values. Mean was used to fill those missing values which may lead to inconsistency and less accuracy in data evaluation.

#### **4.5. Suggestions**

Better algorithms for hyperparameter tuning can be conducted and analyzed to reduce computational cost. More data can be collected to train the model to reduce overfitting and fill the missing values for better accuracy in predicting values.